Facultatea de Automatică şi Calculatoare, UPB
Departamentul de Calculatoare

Analiza algoritmilor

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- Complexitatea algoritmilor (3) -

Metode de rezolvare a recurențelor de complexitate

1. Metoda iterativă
2. Metoda arborelui de recurență
3. Metoda substituției
4. Metoda Master

Notație: $\log_2 n \overset{not}{=} \lg n$

1. Metoda iterativă

Ex 1:

$$T(n) = \begin{cases} K_1, & n = 1 \\ T(\lfloor n/2 \rfloor) + T(\lceil n/2 \rceil) + K_2 n, & n > 1 \end{cases}$$

$$\Rightarrow T(n) = 2T(n/2) + \Theta(n),$$

$$T(1) = \Theta(1)$$

$$T(n) = 2\,T(n/2) + \Theta(n) \quad |\cdot 2^0$$

$$T(n/2) = 2\,T(n/2^2) + \Theta(n/2) \quad |\cdot 2^1$$

$$T(n/2^2) = 2\,T(n/2^3) + \Theta(n/2^2) \quad |\cdot 2^2 \; +$$

$$\dots$$

$$T(n/2^K) = 2\,T(\overbrace{n/2^{K+1}}^{1}) + \Theta(n/2^K) \quad |\cdot 2^K$$

$$T(n) = 2^{K+1} \cdot T(n/2^{K+1}) + \sum_{i=0}^{K} 2^i \cdot \Theta(n/2^i) =$$

$n = 2^{K+1} \Rightarrow K+1 = \lg n$

$K = \lg n - 1$

$$= n \cdot T(1) + (k+1) \cdot \Theta(n) =$$

$$= n \cdot \Theta(1) + \lg n \cdot \Theta(n) =$$

$$= \Theta(n \lg n + n) = \Theta(n \lg n)$$

după ce am aflat clasa de complexitate putem să aducem această clasă de complexitate la o formă mai simplă

Ex 2 :

$$T(n) = \begin{cases} K_1, & n = 1 \\ T(\lfloor n/2 \rfloor) + K_2, & n > 1 \end{cases}$$

$$\Rightarrow T(n) = T(n/2) + \Theta(1),$$

$$T(1) = \Theta(1)$$

$$T(n) = T(n/2) + \Theta(1)$$

$$T(n/2) = T(n/2^2) + \Theta(1)$$

$$\vdots$$

$$T(n/2^K) = T(\underbrace{n/2^{K+1}}_{1}) + \Theta(1)$$

$+$

$$T(n) = T(1) + (K+1)\cdot\Theta(1) =$$

$$= \Theta(1) + \Theta(\lg n) =$$

$$= \Theta(\lg n + 1) = \Theta(\lg n)$$

$n = 2^{K+1}$

$K+1 = \lg n$

$K = \lg n - 1$

Ex 3:

$$T(n) = \begin{cases} K_1, & n = 1 \\ T(n-1) + K_2, & n > 1 \end{cases}$$

$$\Rightarrow \quad T(n) = T(n-1) + \Theta(1),$$

$$T(1) = \Theta(1)$$

$$T(n) = T(n-1) + \Theta(1)$$
$$T(n-1) = T(n-2) + \Theta(1)$$
$$\dots$$
$$T(2) = T(1) + \Theta(1)$$

$+$

$$T(n) = T(1) + (n-1)\cdot\Theta(1) =$$
$$= \Theta(1) + \Theta(n-1) = \Theta(n)$$

2. Metoda arborelui de recurență

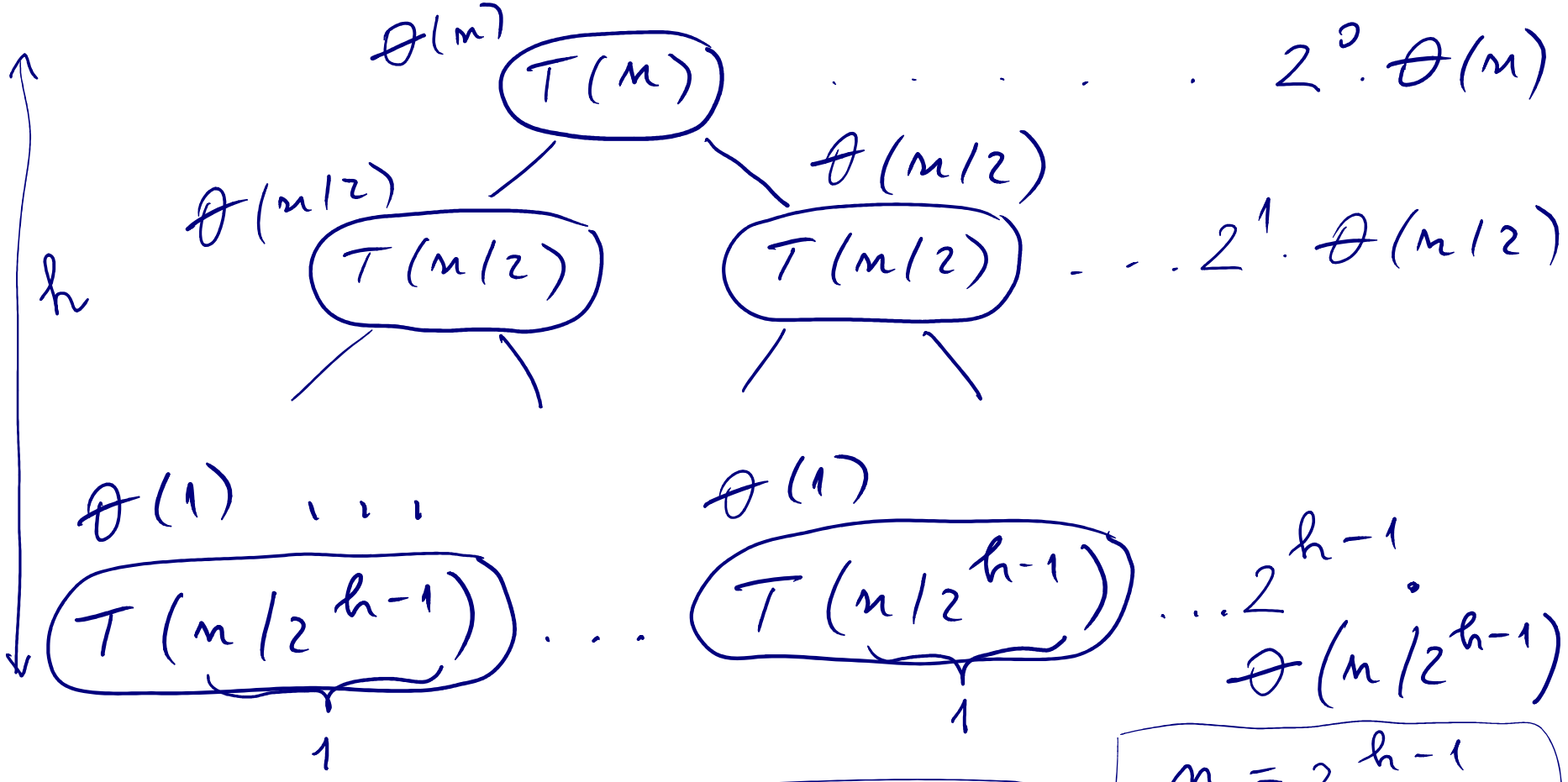
Ex 1:

$$T(n) = \begin{cases} k_1, & n = 1 \\ T(\lfloor n/2 \rfloor) + T(\lceil n/2 \rceil) + k_2 n, & n > 1 \end{cases}$$

$$\Rightarrow T(n) = 2T(n/2) + \Theta(n),$$

$$T(1) = \Theta(1)$$

$$T(n) = 2\,T(n/2) + \Theta(n)\,,\; T(1) = \Theta(1)$$



$$\overset{+}{=\!=\!>}\; T(n) = \sum_{i=0}^{h-1} 2^i \cdot \Theta(n/2^i) =$$

$$n = 2^{h-1}$$
$$h-1 = \lg n$$
$$h = \lg n + 1$$

$$= h \cdot \Theta(n) = (\lg n + 1) \cdot \Theta(n) =$$

$$= \Theta(n \lg n + n) = \Theta(n \lg n)$$

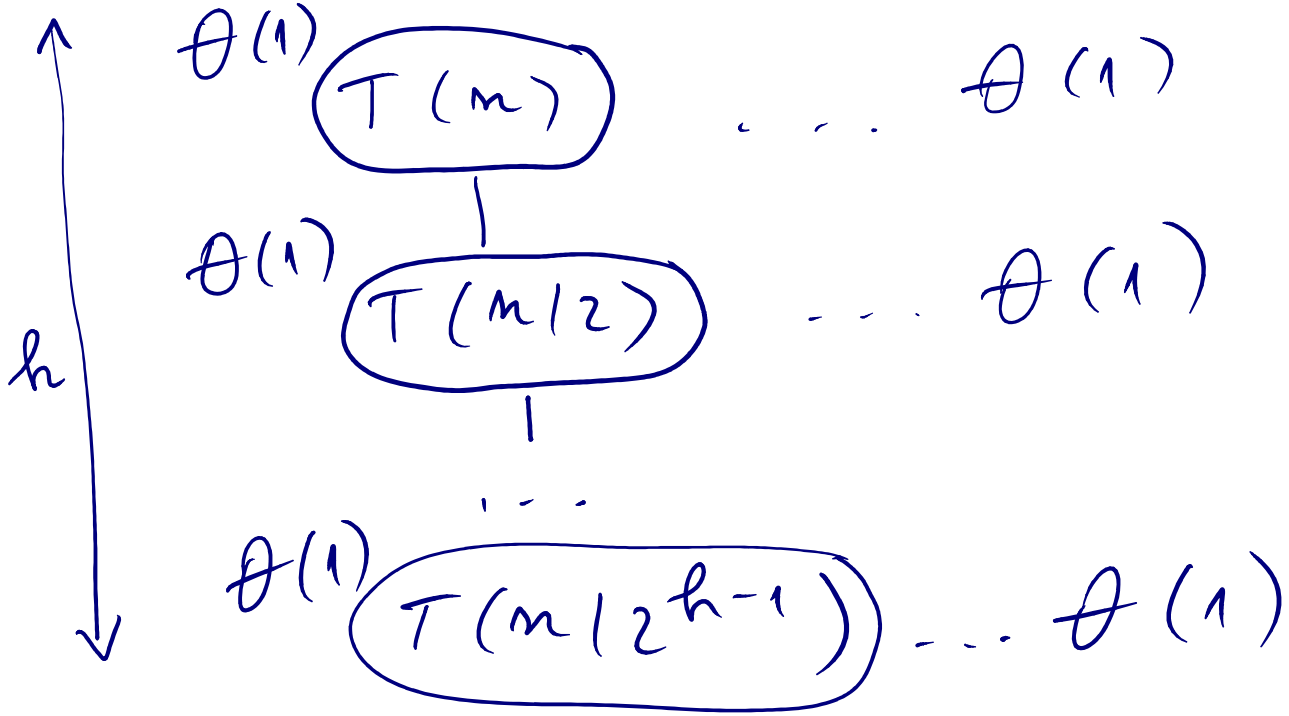
Ex 2 :

$$T(n) = \begin{cases} K_1, & n = 1 \\ T(\lfloor n/2 \rfloor) + K_2, & n > 1 \end{cases}$$

$$\Rightarrow T(n) = T(n/2) + \Theta(1),$$

$$T(1) = \Theta(1)$$

$$T(n) = T(n/2) + \Theta(1), \quad T(1) = \Theta(1)$$



$$n = 2^{h-1}$$
$$h - 1 = \lg n$$
$$h = \lg n + 1$$

$$T(n) = h \cdot \Theta(1) = (\lg n + 1) \cdot \Theta(1) =$$
$$= \Theta(\lg n + 1) = \Theta(\lg n)$$

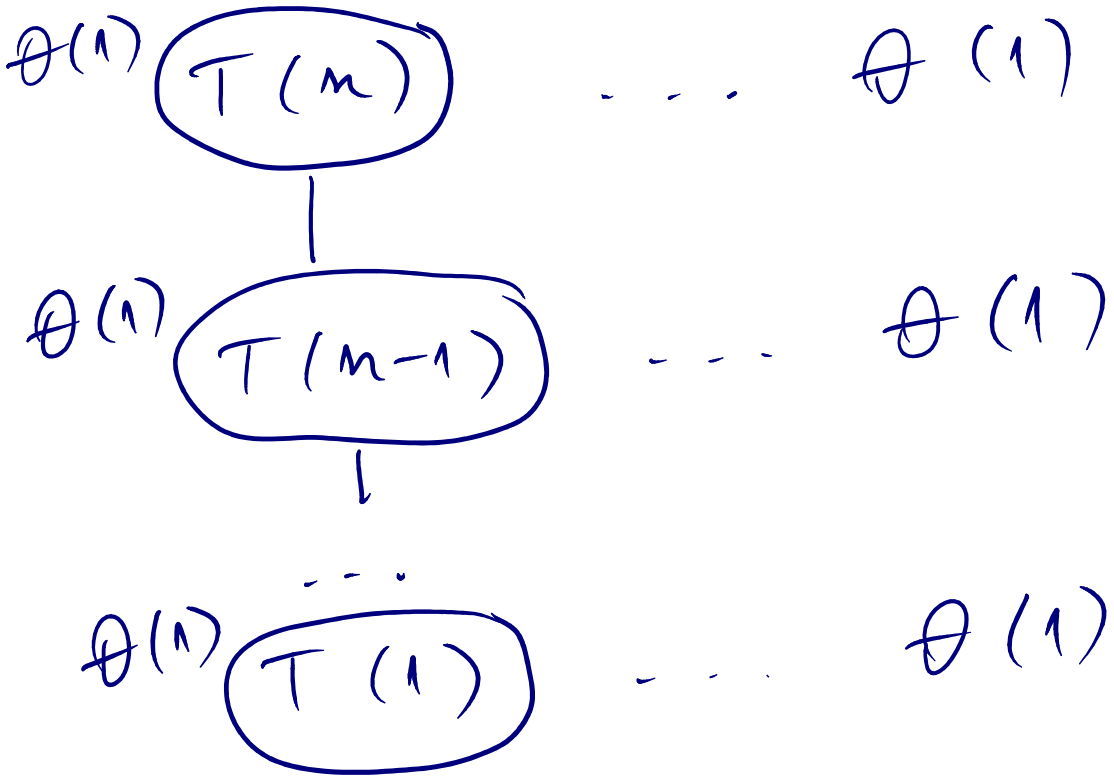
Ex 3 :

$$T(n) = \begin{cases} K_1, & n = 1 \\ T(n-1) + K_2, & n > 1 \end{cases}$$

$$\Rightarrow T(n) = T(n-1) + \Theta(1),$$

$$T(1) = \Theta(1)$$

$$T(n) = T(n-1) + \Theta(1), \quad T(1) = \Theta(1)$$



$$T(n) = n \cdot \Theta(1) = \Theta(n)$$

3. Metoda substituției

- se alege clasa de complexitate (soluția)
- se folosește inducția matematică pentru a afla constantele (pentru clasa de complexitate) și pentru a arăta că soluția verifică formula de recurență.

Ex1 :

$$T(n) = \begin{cases} K_1, & n = 1 \\ T(\lfloor n/2 \rfloor) + T(\lceil n/2 \rceil) + K_2 n, & n > 1 \end{cases}$$

$$\Rightarrow \quad T(n) = 2T(n/2) + K_2 n,$$

$$T(1) = K_1$$

$T(n) = 2T(n/2) + k_2 n$, $T(1) = k_1$

Alegem $T(n) = \Theta(n \lg n + n)$

$\exists c_1, c_2 \in \mathbb{R}_+^*$, $\exists n_0 \in \mathbb{N}^*$ a.î.

$c_1(n \lg n + n) \leq T(n) \leq c_2(n \lg n + n)$,

$\forall n \geq n_0$

Demonstrăm prin inducție inegalitățile de mai sus.

$$T(n) = 2T(n/2) + k_2 n, \quad T(1) = k_1$$

$$c_1(n \lg n + n) \leq T(n) \leq c_2(n \lg n + n)$$

Caz de bază

$n = 1$: $\boxed{c_1 \leq k_1 \leq c_2}$ (1) $\Rightarrow$ $n_0 = 1$

Pas de inducție: $n/2 \rightarrow n$

Ipoteza de inducție:

$$c_1\left(\frac{n}{2} \lg \frac{n}{2} + \frac{n}{2}\right) \leq T\left(\frac{n}{2}\right) \leq c_2\left(\frac{n}{2} \lg \frac{n}{2} + \frac{n}{2}\right)$$

Arătăm că:

$$c_1(n \lg n + n) \leq T(n) \leq c_2(n \lg n + n)$$

$$T(n) = 2T(n/2) + k_2 n \quad , T(1) = k_1$$

$$c_1(n \lg n + n) \le T(n) \le c_2(n \lg n + n)$$

$$c_1\left(\frac{n}{2}\lg\frac{n}{2} + \frac{n}{2}\right) \le T\left(\frac{n}{2}\right) \le c_2\left(\frac{n}{2}\lg\frac{n}{2} + \frac{n}{2}\right)$$

$$2c_1\left(\frac{n}{2}\lg\frac{n}{2} + \frac{n}{2}\right) + k_2 n \le 2T\left(\frac{n}{2}\right) + k_2 n \le$$

$$\le 2c_2\left(\frac{n}{2}\lg\frac{n}{2} + \frac{n}{2}\right) + k_2 n$$

$$c_1 n \lg\frac{n}{2} + c_1 n + k_2 n \le T(n) \le$$

$$\le c_2 n \lg\frac{n}{2} + c_2 n + k_2 n$$

$$c_1 n \lg n - c_1 n + c_1 n + k_2 n \le T(n) \le$$

$$\le c_2 n \lg n - c_2 n + c_2 n + k_2 n$$

$$c_1(n\lg n + n) + \underbrace{(k_2 - c_1)}_{\geq 0} \cdot n \leq T(n) \leq$$

$$c_1(n\lg n + n \leq$$

$$\leq c_2(n\lg n + n) + \underbrace{(k_2 - c_2)}_{\leq 0} \cdot n$$

$$\leq c_2(n\lg n + n)$$

$$\Rightarrow \begin{cases} k_2 \geq c_1 \\ k_2 \leq c_2 \end{cases} \Rightarrow \boxed{c_1 \leq k_2 \leq c_2} \quad (2)$$

Caz de bază : $c_1 \leq k_1 \leq c_2$

$n_0 = 1$

Pas de inducție : $c_1 \leq k_2 \leq c_2$

$\Rightarrow \exists c_1 = \min(k_1, k_2), c_2 = \max(k_1, k_2) \in \mathbb{R}_+^*$,

$\exists n_0 = 1 \in \mathbb{N}^*$ a.î.

$c_1(n \lg n + n) \leq T(n) \leq c_2(n \lg n + n)$,

$\forall n \geq n_0$

$\Rightarrow \boxed{T(n) = \Theta(n \lg n + n)}$

Ex 2 :

$$T(n) = \begin{cases} K_1, n = 1 \\ T(n-1) + K_2, n > 1 \end{cases}$$

$$\Rightarrow T(n) = T(n-1) + K_2,$$

$$T(1) = K_1$$

$T(n) = T(n-1) + k_2,\ T(1) = k_1$

Alegem $T(n) = \Theta(n)$

$\Rightarrow \exists c_1, c_2 \in \mathbb{R}_+^*, \exists n_0 \in \mathbb{N}^*$ a.î.

$$c_1 n \leq T(n) \leq c_2 n, \forall n \geq n_0$$

Demonstrăm prin inducție inegalitățile de mai sus.

Caz de bază

$n = 1$: $\boxed{c_1 \leq k_1 \leq c_2}$ (1) $\Rightarrow n_0 = 1$

Pas de inducție : $n-1 \rightarrow n$

ipoteza de inducție

$c_1 (n-1) \leq T(n-1) \leq c_2 (n-1)$

Arătăm că

$c_1 n \leq T(n) \leq c_2 n$

$c_1 (n-1) \leq T(n-1) \leq c_2 (n-1) \mid + K_2$

$c_1 (n-1) + K_2 \leq T(n-1) + K_2 \leq$

$\leq c_2 (n-1) + K_2$

$c_1 (n-1) + K_2 \leq T(n) \leq c_2 (n-1) + K_2$

$$c_1(n-1) + k_2 \leq T(n) \leq c_2(n-1) + k_2$$

$$c_1 n + k_2 - c_1 \leq T(n) \leq c_2 n + k_2 - c_2$$

$k_2 - c_1 \geq 0$

$k_2 - c_2 \leq 0$

$c_1 n \leq$

$\leq c_2 n$

$$\Rightarrow \begin{cases} k_2 \geq c_1 \\ k_2 \leq c_2 \end{cases} \Rightarrow \boxed{c_1 \leq k_2 \leq c_2} \quad (2)$$

Caz de bază : $c_1 \leq k_1 \leq c_2$

$n_0 = 1$

Pas de inductie : $c_1 \leq k_2 \leq c_2$

$\Rightarrow \exists c_1 = \min(k_1, k_2),\ c_2 = \max(k_1, k_2) \in \mathbb{R}_+^*,$

$\exists n_0 = 1 \in \mathbb{N}^*$ a.î.

$c_1 n \leq T(n) \leq c_2 n,\ \forall n \geq n_0$

$\Rightarrow \boxed{T(n) = \Theta(n)}$